

CLIP

DATA 443/643 · Week 5, note 5.1

Week 5

American University

Ahmad Mousavi

Ideas from Stanford CS231N 2025 L16 (two towers, a cosine, not a captioner). Original slides; that course is not copied.

Lecture plan

- 1 Two towers; keep both after training
- 2 A similarity score, not a sentence
- 3 A 2 by 2 cosine matrix
- 4 Temperature and batch size

Key goal: leave able to write `encode_image`, `encode_text`, `cosine` — and to refuse generate.

First time: what / why / formula

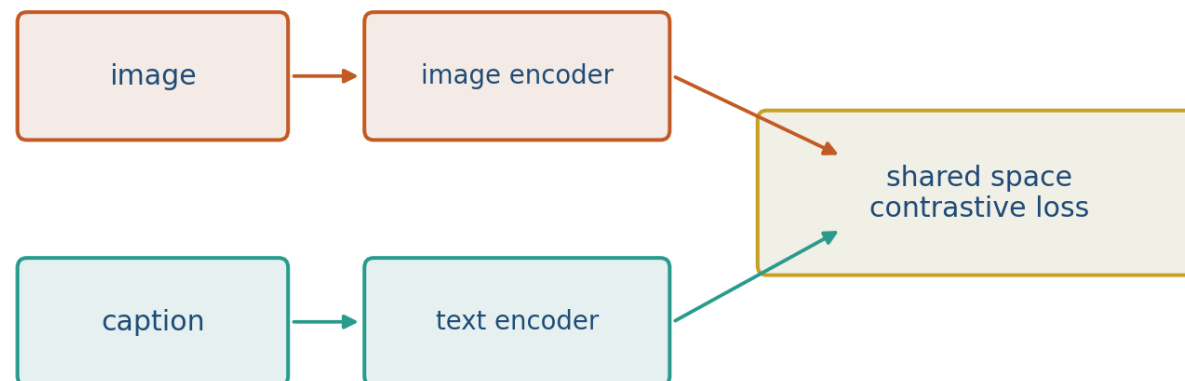
- What / why. Image encoder + text encoder, InfoNCE on the batch, keep both towers. One model that retrieves and zero-shot classifies without a detector or a decoder.
- Architecture. ``encode_image``, ``encode_text``, cosine. No ``generate``.
- Formula. Same InfoNCE as 4.3 with learned τ . API is three calls.
- Tradeoffs. + Zero-shot and retrieval. – Not a captioner; hungry for batch size; prompt wording is method; bias next note.

How. Train on web pairs. At test, embed the image and a list of prompts. Note 5.2 adds a decoder (BLIP).

Two towers, one cosine

- ViT or ResNet for the image. A transformer for the text.
- InfoNCE on the batch (note 4.3).
- After training you keep the towers, not a fused head.

CLIP: two towers, one cosine, a batch of matches



What the API is allowed to do

CLIP

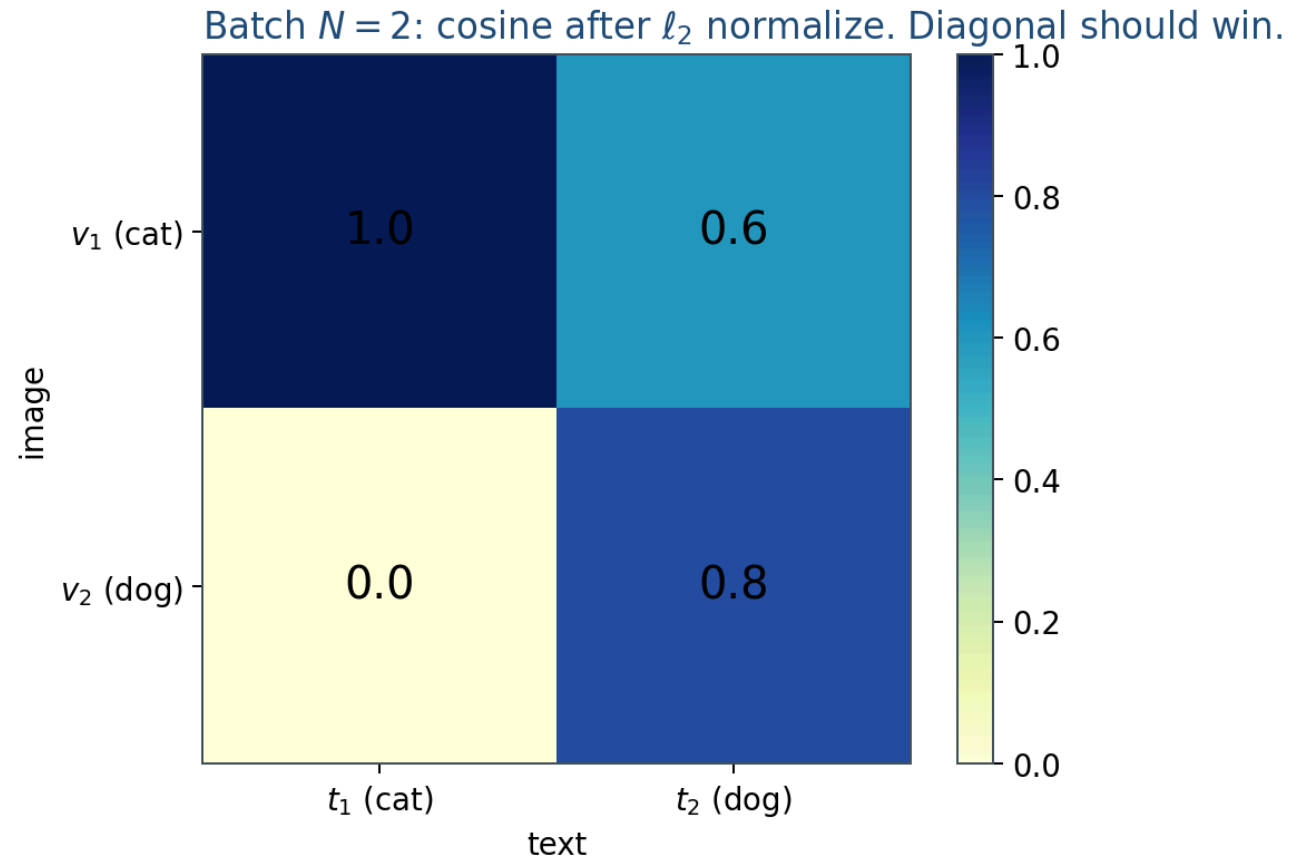
Score a pair. Retrieve. Zero-shot by nearest prompt.
Freeze the image tower later.

Not CLIP

Decode a caption. Draw boxes. Guarantee the photo is true. That is BLIP, a detector, or a human.

A batch of 2 you will write

- Unit vectors. $S_{11} = 1.0$, $S_{12} = 0.6$, $S_{21} = 0$, $S_{22} = 0.8$.
- The diagonal should win each row.
- One negative is not CLIP scale. It is the algebra.



InfoNCE on row 1

$$-\log \frac{\exp(S_{11} / \tau)}{\exp(S_{11} / \tau) + \exp(S_{12} / \tau)}$$

$\tau = 1$: softmax about [0.60, 0.40],
NLL about 0.51.

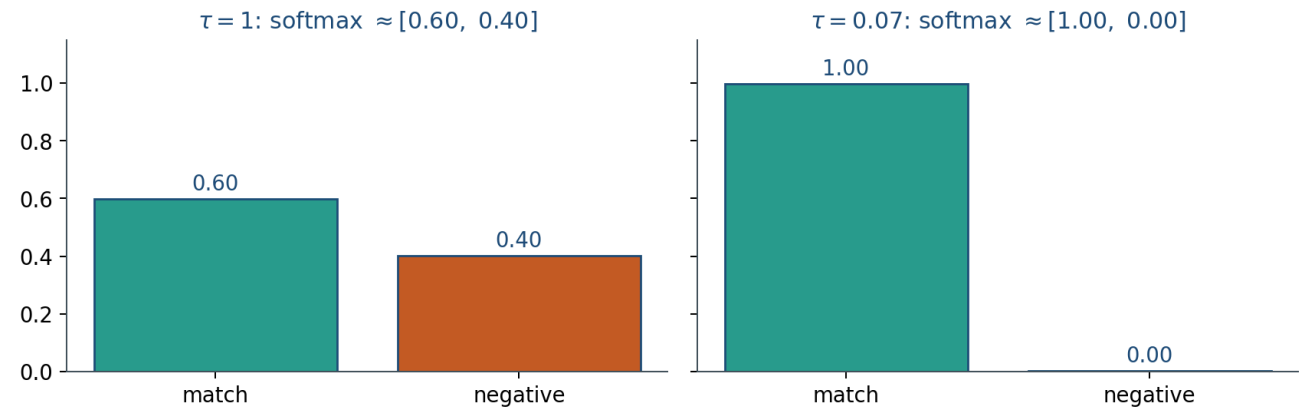
$\tau = 0.07$: match probability
about 0.997.

Huge batches exist so you have many
negatives.

Small tau, large batch

- Learned τ is often near 0.07.
- A batch of 8 has 7 negatives. A batch of 1024 has 1023.
- Fine-grained word order still needs hard negatives later.

Same scores [1.0, 0.6]. Small τ makes the one negative brutal.



What to take from this lecture

Three calls

encode_image, encode_text,
cosine. No generate.

Temperature

Same scores, different softmax.
Check unit length first.

Next

If you must write a sentence, that is
5.2.